Software Project Management

Assignment 3

Module 6: Communication

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Table of Contents

1. Assignment Objective

In continuation of the semester project, this assignment focuses on developing cost estimates and a project budget for the Communication Module. Since this is a university-based project, all financial values are assumed based on realistic market rates and logical reasoning.

2. Part 1: Identification of Cost Components

2.1 Cost Components Table

Category	Cost Type	Responsibilities/Description	Assumptions
Human Resource Costs	Backend Developers	Implement WebSocket messaging, encryption, multimedia sharing, message notification, storage in database.	3 developers working part-time PKR 2700/day
	Frontend Developer	Design communication interface, integrate backend with UI	2 designers working part-time PKR 2620/day
	QA/Testers	Test real time messaging, encryption, notifications, message received to correct user, file properly shared.	1 tester PKR 2660/day
	Team Lead	Coordination, requirement tracking, meetings	1 person - shared role as developer PKR 3920/day
Software and Tools Costs	GitHub	Version control	Free

	Development Tools (VS Code)	Coding	Free
Infrastructure Costs	WebSocket Hosting	For real-time messaging	Free-tier hosting
	Database storage	Media (File, Image, Video) storage and message history	No cost, local database
	Daily.co API	Video Meeting Links	PKR 7500 per month
	Push Notifications	Show message alert on the app	No cost
Security Costs	End-to-End Encryption	Implemented using libraries	Open-source
Communication & Coordination Costs	Internet / Meetings	Team collaboration via google meet	No cost
Miscellaneous	Misc Expenses	Minor expenses	PKR 5,000
Contingency	Risk buffer	Standard PM practice	10%
Management Reserve	Unknown risks	Standard PM practice	5%

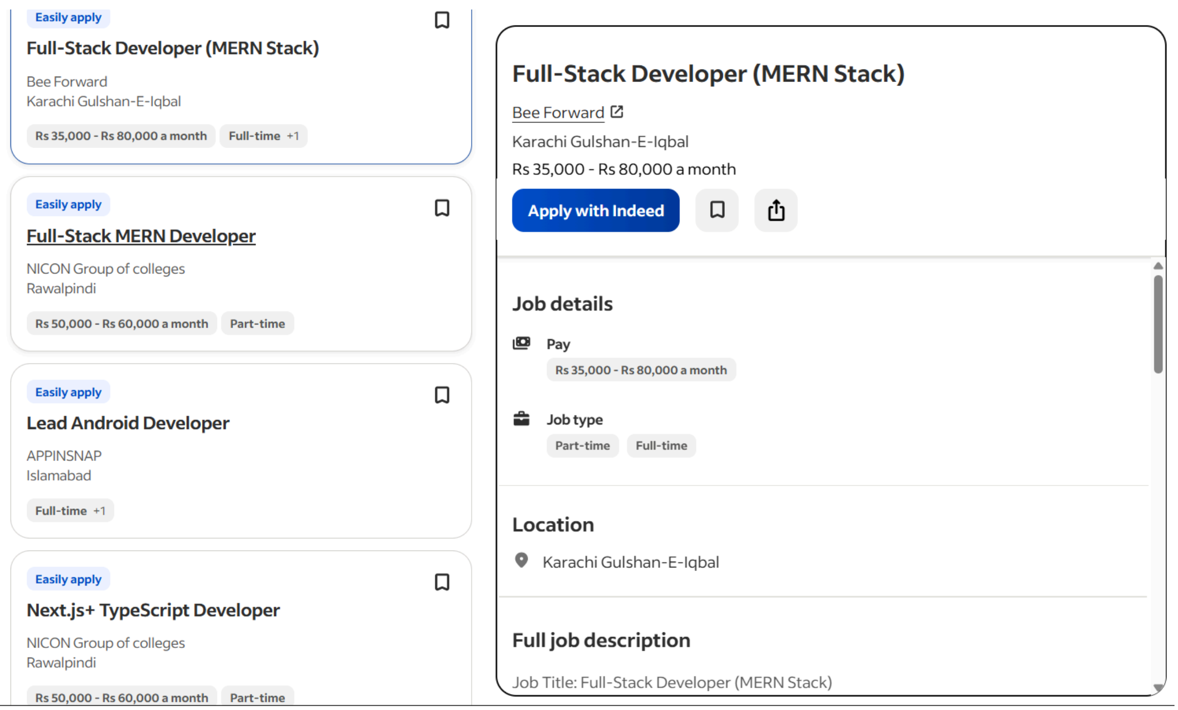
2.2 Assumptions and Justification

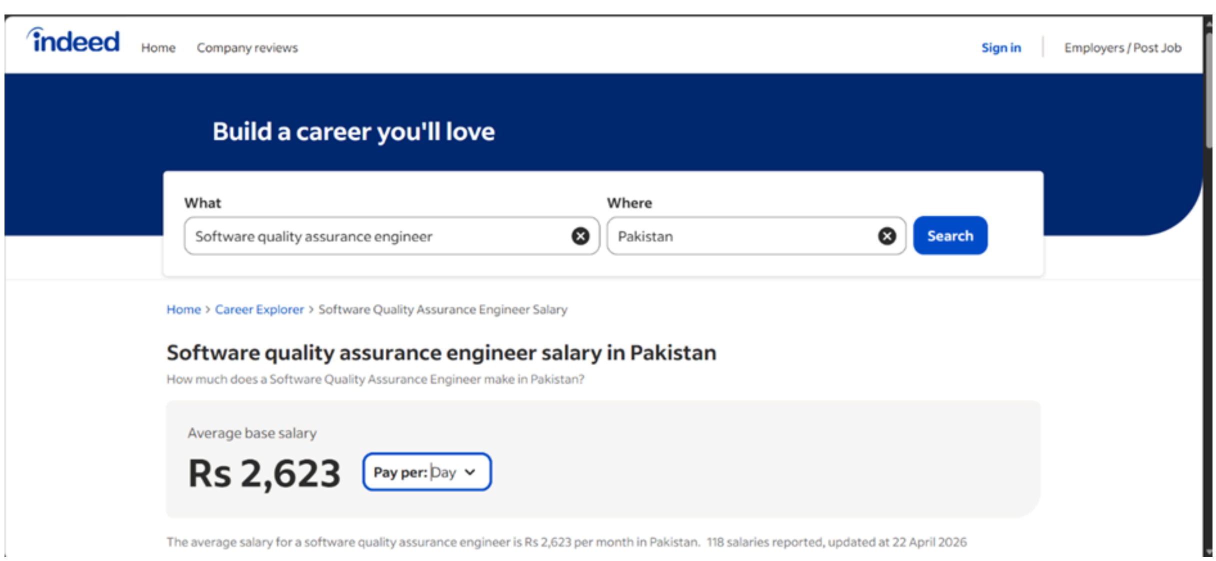
2.2.1 Human Resource Costs:

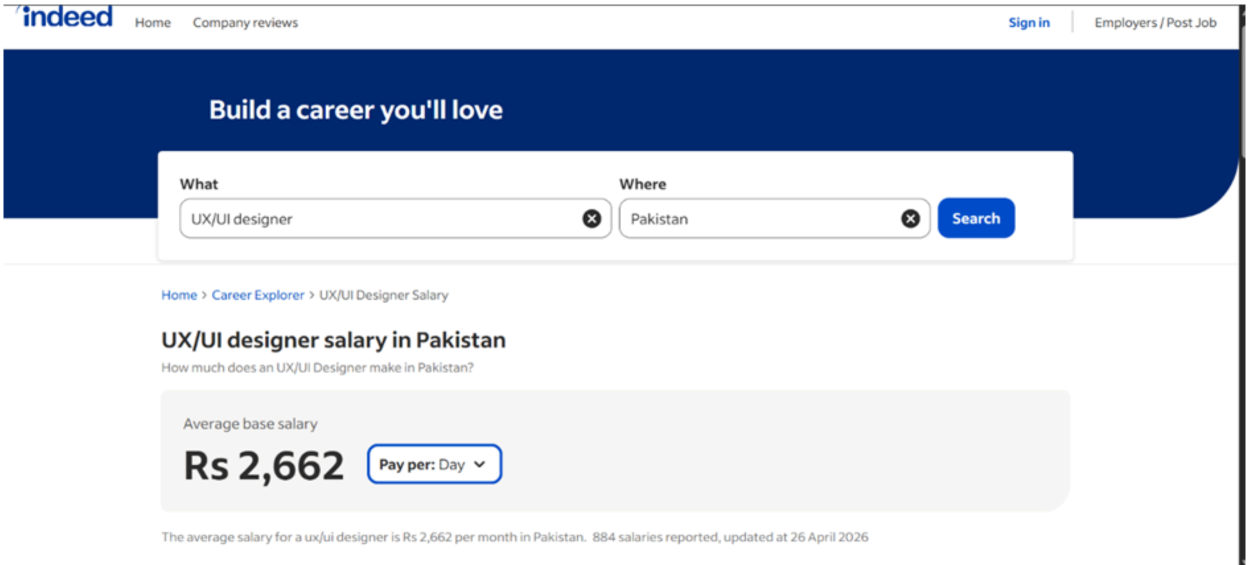
Assumptions:

- Students are implementing the system, but market-equivalent salaries are used.
- Rates are based on entry-level software industry standards in Pakistan and according to our skill sets.

Justification:



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Build a career you'll love
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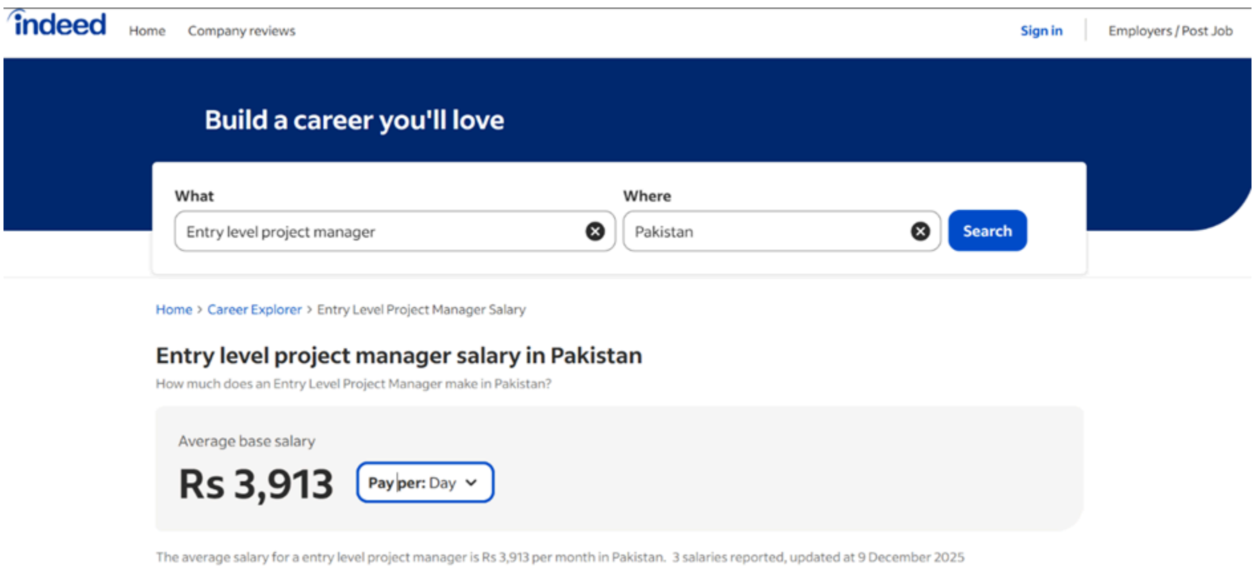
UX/UI designer salary in Pakistan
How much does an UX/UI Designer make in Pakistan?
Average base salary
Rs 2,662
Pay per: Day
The average salary for a ux/ui designer is Rs 2,662 per month in Pakistan. 884 salaries reported, updated at 26 April 2026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Where
Pakistan
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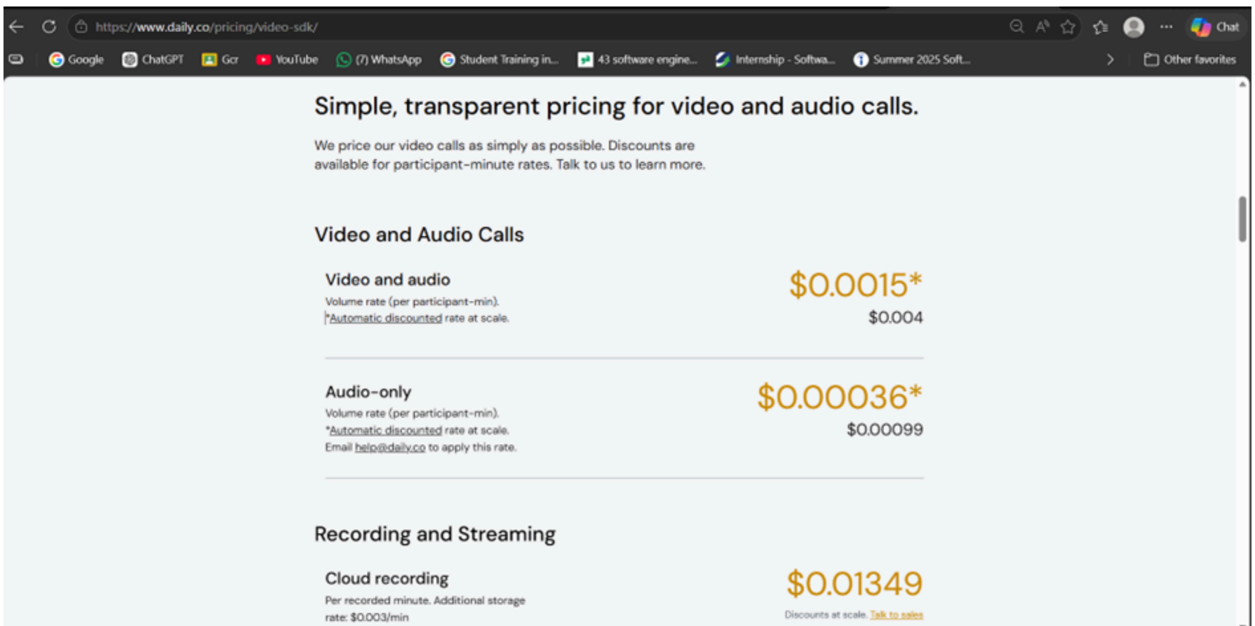
Home > Career Explorer > Entry Level Project Manager Salary
Entry level project manager salary in Pakistan
How much does an Entry Level Project Manager make in Pakistan?
Average base salary
Rs 3,913
Pay per: Day
The average salary for a entry level project manager is Rs 3,913 per month in Pakistan. 3 salaries reported, updated at 9 December 2025

2.2.2 Daily.co API (Video Meeting Links)

Assumptions:

- Estimated at 0.9 USD/day ≈ PKR 276/day
- Monthly ≈ PKR 7,500
- Based on assumed API usage for video meetings.

Justification:



3. Part 2: Cost Estimation Table

3.1 Basis of Cost Estimation

3.1.1 Currency and Monitoring Unit

All project costs are estimated in Pakistani Rupees (PKR).
The primary monitoring units used for estimation are:

- Hours for effort estimation
- Resource costs were converted into hourly rates using an 8-hour working day assumption.

3.1.2 Estimation Approach

The Communication Module cost estimation follows the Bottom-Up Estimation approach using a Work Breakdown Structure (WBS).

In this approach:

- The project was divided into smaller deliverables and activities.
- Each task was individually estimated in terms of required effort (hours/days).
- The effort was multiplied by the assigned resource rate to calculate task cost.
- Individual task costs were aggregated to calculate work package costs and the total project budget.

This estimation technique was selected because it:

- Provides accurate task-level estimation
- Improves cost visibility and tracking
- Helps allocate resources efficiently
- Supports better planning for real-time communication features

3.1.3 Role Definitions and Rates

Role	Monthly Salary (PKR)	Working Days/Month	Daily Rate (PKR)	Hourly Rate (PKR)	Source
Team Lead (Amna)	PKR 102,000	26 days	PKR 3,920	PKR 490	Entry-level project manager rate on Indeed.pk (2026)
Backend Developers (Amna + Haifa)	PKR 70,200	26 days	PKR 2,700	PKR 337.5	Estimated backend developer (MERN) salary on Indeed.pk (2026)
QA / Tester (Nisha)	PKR 69,160	26 days	PKR 2,620	PKR 332.5	SQA engineer salary on Indeed.pk (22 April, 2026)
UI/UX Designer (Tanees)	PKR 68,120	26 days	PKR 2,660	PKR 327.5	Estimated UI/UX designer salary on Indeed.pk (26 April, 2026)
Communication Module Team Average	PKR 77,370	26 days	PKR 2,975	PKR 372	Calculated blended average of all team roles

Hourly rates were derived from entry-level/part-time software industry salaries in Pakistan.

3.1.4 Understanding of Collaborative Cost Calculation

Some activities in the Communication module were completed collaboratively by multiple roles instead of a single resource. In such cases, the hourly cost rate was calculated by taking the average of the participating roles' hourly rates.

This approach was used because tasks such as requirement analysis, documentation, testing, and UI planning required combined contributions from technical and management resources. Examples include:

Collaborative Roles	Formula	Average Hourly Rate
Team lead + Developer	(490+337.5)/2 = 827.5/2	PKR 414/hour
Developer + UI/UX	(337.5+332.5)/2 = 670/2	PKR 335/hour
Developer + QA	(337.5+327.5)/2 = 665/2	PKR 332.5/hour
All team members	-	PKR 500/hour

These average rates were applied in the cost breakdown table wherever collaborative work was involved.

3.1.5 Assumptions

The following assumptions were made during project cost estimation:

1. The project duration is estimated as 2 months (approximately 54 working days).
2. The system is developed in an academic environment by students, however, market-equivalent entry-level salaries were taken to estimate realistic project costs and to convert effort into realistic monetary values.
3. Open-source technologies and tools like node.js, express.js, postgres, etc are used; therefore, no licensing costs or software purchase costs were included.
4. No laptops, smartphones, microphones, or internet devices were budgeted because team members are assumed to have used their personal resources.
5. Daily.co API pricing was estimated based on assumed meeting usage frequency.
6. Communication and collaboration costs are assumed minimal.
7. Real-time communication features may require additional debugging and testing effort.

3.1.5 Constraints and Risks

Constraints

The project estimation was prepared under the following constraints:

- Limited academic project duration
- Part-time student development availability
- Dependence on free/open-source technologies
- Limited infrastructure budget
- No dedicated hardware or paid cloud hosting

Risks

The Communication Module contains technically sensitive components that may increase project uncertainty, including:

- Real-time WebSocket synchronization issues
- API integration failures
- Encryption and media-sharing complexity
- Notification delivery delays
- Database performance bottlenecks
- Unexpected debugging and testing effort
- Schedule overruns due to integration complexity

3.1 Project Cost Estimate

WBS Items	# Units/ Hrs	Cost/Unit/Hr	Subtotals	WBS Level 2 Totals	% of Total
1. Project Management				**PKR 12,844**	**6.57%**
6.1.1 Project Plan	10	490	PKR 4,900		
6.1.2 Meeting Records	6	490	PKR 2,944		
6.1.3 Progress Reports	10	500	PKR 5,000		
2. Requirements & Design				**PKR 12,140**	**6.21%**
6.2.1 SP-SRS	4	500	PKR 2,000		
6.2.2 Requirements Analysis	12	600	PKR 7,200		
6.2.3 Requirements Documentation	6	490	PKR 2,940		
2. System Design				**PKR 19,552**	**10.53%**
6.3.1 System Architecture	18	414	PKR 7,452		
6.3.2 Database Schema	8	337.5	PKR 2,700		
6.3.3 API Design	8	337.5	PKR 2,700		
6.3.4 UI/UX Design	20	335	PKR 6,700		
3. Software Development*				**PKR 68,066**	**34.79%**
6.4.1 Messaging System	30	500	PKR 15,000		
6.4.2 Message Sync	8	414	PKR 3,312		
6.5.1 Group Management	8	414	PKR 3,312		
6.5.2 Group Messaging	18	414	PKR 7,452		
6.6.1 File Sharing	22	374	PKR 8,228		
6.6.2 Secure Storage	4	337.5	PKR 1,350		
6.7.1 Notification System	8	335	PKR 2,680		
6.7.2 Notification Trigger	4	337.5	PKR 1,350		

6.8.1 Link Generation	8	374	PKR 2,992		
6.8.2 Link Sharing	14	332.5	PKR 4,655		
6.9.1 Message Storage	6	337.5	PKR 2,025		
6.9.2 Chat Retrieval	4	337.5	PKR 1,350		
6.10.1 Chat UI	16	327.5	PKR 5,240		
6.10.2 Notification UI	8	327.5	PKR 2,620		
6.10.3 Interaction Flows	13	500	PKR 6,500		
4. Integration				**PKR 22,333**	**11.41%**
6.11.1 Messaging APIs	21	414	PKR 8,695		
6.11.2 User Integration	5	337.5	PKR 1,638		
6.11.3 Module Integration	24	500	PKR 12,000		
5. Testing & QA				**PKR 16,625**	**8.50%**
6.12.1 Unit Testing	8	332.5	PKR 2,660		
6.12.2 Integration Testing	16	332.5	PKR 5,320		
6.12.3 System Testing	16	332.5	PKR 5,320		
6.12.4 Bug Logs	10	327.5	PKR 3,325		
6. Documentation				**PKR 11,500**	**5.87%**
6.13.1 SP-SRS Doc	4	500	PKR 2,000		
6.13.2 SP-SDS Doc	4	500	PKR 2,000		
6.13.3 Technical Docs	15	500	PKR 7,500		
7. Deployment & Closeout				**PKR 33,550**	**17.15%**
6.14.1 Deployment	4	337.5	PKR 1,350		
6.14.2 Demonstration	4	500	PKR 7,200		
6.14.3 Final Report	10	500	PKR 5,000		
Daily.co API (2 months)	2 months	7,500	PKR 15,000		
Miscellaneous Expenses	-	-	PKR 5,000		

Subtotal (Direct Costs)				**PKR 195,610**	
Contingency Reserve (10% of subtotal)	-	-	PKR 19,561	**PKR 19,561**	10%
Management Reserve (5% of subtotal)	-	-	PKR 9780.5	**PKR 9780**	5%
Total Project Cost Estimate				**PKR 224,951**	

3.2 Software Development Estimate

1. LABOUR ESTIMATE				
Role/Category	**# Units/Hr**	**Cost/Unit/Hr**	**Subtotals**	**Calculations**
Backend Developer labour estimate (2)	200	337.5	PKR 67,500	200 × 337.5
QA/Tester labour estimate (1)	94	332.5	PKR 31,255	94 × 332.5
Frontend Developer labour estimate (1)	80	327.5	PKR 26,200	80 × 327.5
Total labor estimate			**PKR 124,955**	Sum above 3 values
2. FUNCTION POINT ESTIMATE				
Component Type	**Quantity**	**Conversion Factor**	**Function Points**	**Calculations**
External inputs (message input, file upload, group creation)	8	4	32	8 × 4
External interface files (API integrations, file formats)	4	6	48	4 × 6
External outputs (notifications, chat display, file display)	7	5	35	7 × 5
External queries (message search, chat history retrieval)	10	4	40	10 × 4
Logical internal tables (user relationships, message store, file metadata)	8	8	64	8 × 8
Total function points			**219**	Sum above values

Typescript/Node.js language equivalency value	50			Assumed academic reference value
Source lines of code (SLOC) estimate			**PKR 10,950**	219 × 50
Productivity × KSLOC ^ Penalty (in months)			**21.9**	2 × 10.95^ 1.0
Total labor hours			**416**	1.9 × 219
Cost/labor hour			**PKR 370**	Blended average hourly rate
Total function point estimate			**PKR 153,920**	416 × 370

4. Part 3: Creating Time-Phased Budget

4.1 Time-Phased Budget

WBS Items	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Totals
6.1 Project Management	5,500	3,500	2,000	1,844	-	-	-	-	12,844
6.1.1 Project Plan	4,900	-	-	-	-	-	-	-	4,900
6.1.2 Meeting Records	300	1,200	744	700	-	-	-	-	2,944
6.1.3 Progress Reports	300	2,300	1,256	1,144	-	-	-	-	5,000
6.2 Requirements & Design	6,000	6,140	-	-	-	-	-	-	12,140
6.2.1 SP-SRS	2,000	-	-	-	-	-	-	-	2,000
6.2.2 Requirements Analysis	3,500	3,700	-	-	-	-	-	-	7,200
6.2.3 Requirements Documentation	500	2,440	-	-	-	-	-	-	2,940
6.3 System Design	-	8,000	11,552	-	-	-	-	-	19,552
6.3.1 System Architecture	-	4,000	3,452	-	-	-	-	-	7,452
6.3.2 Database Schema	-	1,500	1,200	-	-	-	-	-	2,700
6.3.3 API Design	-	1,500	1,200	-	-	-	-	-	2,700
6.3.4 UI/UX Design	-	1,000	5,700	-	-	-	-	-	6,700

6.4–6.10 Software Development	-	-	12,000	14,000	16,000	12,000	8,066	6,000	68,066
6.4.1 Messaging System	-	-	5,000	5,000	5,000	-	-	-	15,000
6.4.2 Message Sync	-	-	1,500	1,812	-	-	-	-	3,312
6.5.1 Group Management	-	-	1,500	1,812	-	-	-	-	3,312
6.5.2 Group Messaging	-	-	2,000	3,000	2,452	-	-	-	7,452
6.6.1 File Sharing	-	-	-	3,000	3,228	2,000	-	-	8,228
6.6.2 Secure Storage	-	-	-	1,350	-	-	-	-	1,350
6.7.1 Notification System	-	-	-	1,000	1,680	-	-	-	2,680
6.7.2 Notification Trigger	-	-	-	1,350	-	-	-	-	1,350
6.8.1 Link Generation	-	-	-	1,500	1,492	-	-	-	2,992
6.8.2 Link Sharing	-	-	-	-	2,655	2,000	-	-	4,655
6.9.1 Message Storage	-	-	-	-	1,025	1,000	-	-	2,025
6.9.2 Chat Retrieval	-	-	-	-	1,350	-	-	-	1,350
6.10.1 Chat UI	-	-	2,000	-	1,240	2,000	-	-	5,240
6.10.2 Notification UI	-	-	-	-	1,620	1,000	-	-	2,620
6.10.3 Interaction Flows	-	-	-	-	2,258	2,242	2,000	-	6,500
6.11 Integration	-	-	-	-	-	8,000	8,333	6,000	22,333
6.11.1 Messaging APIs	-	-	-	-	-	4,000	2,695	2,000	8,695

6.11.2 User Integration	-	-	-	-	-	1,638	-	-	1,638
6.11.3 Module Integration	-	-	-	-	-	2,362	5,638	4,000	12,000
6.12 Testing & QA	-	-	-	-	-	4,000	6,625	6,000	16,625
6.12.1 Unit Testing	-	-	-	-	-	2,660	-	-	2,660
6.12.2 Integration Testing	-	-	-	-	-	1,340	3,980	-	5,320
6.12.3 System Testing	-	-	-	-	-	-	2,660	2,660	5,320
6.12.4 Bug Logs	-	-	-	-	-	-	3,325	-	3,325
6.13 Documentation	-	-	-	-	-	-	5,000	6,500	11,500
6.13.1 SP-SRS Doc	-	-	-	-	-	-	2,000	-	2,000
6.13.2 SP-SDS Doc	-	-	-	-	-	-	2,000	-	2,000
6.13.3 Technical Docs	-	-	-	-	-	-	1,000	6,500	7,500
6.14 Deployment & Closeout	-	-	-	-	-	-	12,000	21,550	33,550
6.14.1 Deployment	-	-	-	-	-	-	1,350	-	1,350
6.14.2 Demonstration	-	-	-	-	-	-	5,000	2,200	7,200
6.14.3 Final Report	-	-	-	-	-	-	2,000	3,000	5,000
Daily.co API (2 months)	-	-	-	-	-	-	3,000	12,000	15,000
Miscellaneous Expenses	-	-	-	-	-	-	650	4,350	5,000
Subtotal (Direct Costs)	**11,500**	**17,640**	**25,552**	**15,844**	**16,000**	**24,000**	**40,368**	**44,706**	**195,610**

Contingency Reserve (10%)	-	-	-	-	-	-	9,561	10,000	19,561
Management Reserve (5%)	-	-	-	-	-	-	4,780	5,000	9,780
TOTALS	**11,500**	**17,640**	**25,552**	**15,844**	**16,000**	**24,000**	**54,709**	**59,706**	**224,951**

4.2 Project Budget Components Summary

Work Package Cost Estimates (Cost Baseline)	**PKR 195,610**
Contingency Reserve	**PKR 19,561**
Management Reserve	**PKR 9,780**
Total Project Budget	**PKR 224,951**

4.3 Budget Baseline Explanation

The above table represents the time-phased budget baseline for the **Communication Module of the National Freelance and Skill Verification Platform**. The project budget has been distributed across an **eight-week development schedule** to show how project costs are expected to be incurred over time. This approach ensures consistency between the project scope, work breakdown structure (WBS), development schedule, and estimated costs.

During the initial weeks (Week 1 and Week 2), the majority of the cost is associated with project management, requirements gathering, SRS preparation, and system design activities. These activities establish the project foundation and therefore occur at the start of the project lifecycle.

From Week 3 to Week 6, the project enters the major development phase. The highest portion of the budget is allocated during these weeks because core functionalities such as:

- real-time messaging,
- file sharing,
- encryption,
- notification handling,

- meeting integration,
- and database message history

are implemented during this stage. Development activities require the highest resource utilization; therefore, the budget expenditure increases significantly during these weeks.

In the later weeks (Week 7 and Week 8), costs shift toward:

- integration,
- quality assurance,
- system testing,
- bug fixing,
- technical documentation,
- deployment,
- and final reporting.

Reserve allocations are also placed near the final phase of the project because contingency and management reserves are generally utilized only if additional effort, debugging, or risk handling becomes necessary during implementation.

The time-phased budget baseline helps monitor planned spending throughout the project lifecycle and provides a clear view of when financial resources are expected to be consumed. This improves project tracking, cost control, and schedule management for the Communication Module.

4.4 Project Budget Components Explanation

The total project budget for the Communication Module is composed of multiple layered components, structured as follows:

4.4.1 Activity Cost Estimates

These represent the lowest-level cost estimates derived from individual WBS activities such as real-time messaging, file sharing, testing, and UI design. Each activity cost is calculated based on effort (hours/days) multiplied by resource rates.

4.4.2 Work Package Cost Estimates

Activity costs are grouped into work packages such as:

- Development
- Testing
- Design
- Documentation

These form the primary building blocks of the project cost.

4.4.3 Contingency Reserve

A **contingency reserve of 10%** is added to account for identified risks such as:

- real-time system failures,
- API integration issues,
- delays in testing or debugging.

This reserve is included within the cost baseline.

4.4.4 Management Reserve

An additional **5% management reserve** is included to handle unforeseen risks that cannot be predicted during planning. This reserve is not part of the cost baseline but is included in the total project budget.

4.4.5 Cost Baseline

The cost baseline is the sum of:

- all work package costs
- plus contingency reserve

It represents the planned spending over time and is reflected in the time-phased budget table.

4.4.6 Total Project Budget

The final project budget is:

Total Budget = Cost Baseline + Management Reserve + Contingency Reserve

For the Communication Module:

Total Budget = PKR 224,951 = PKR 225,000